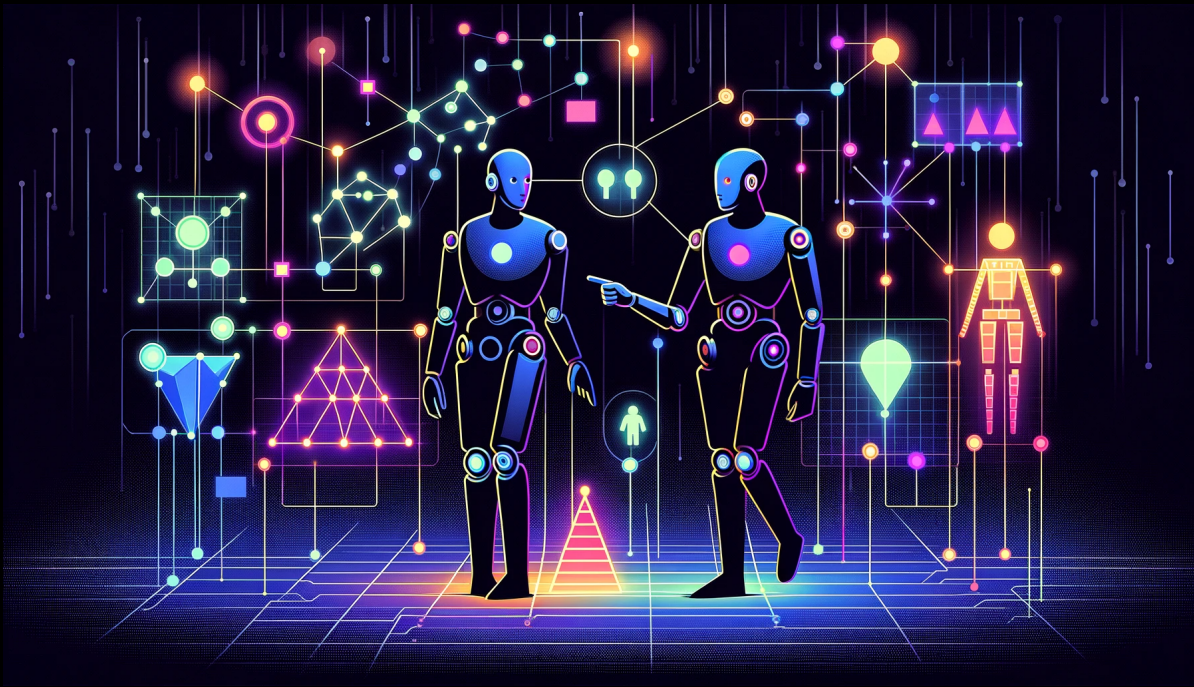


Exploring Language Compositionality in Multi-Agent Systems



Mees A. Lindeman

Layout: typeset by the author using L^AT_EX.

Cover illustration: Generated by OpenAI's DALL·E

Exploring Language Compositionality in Multi-Agent Systems

Mimicking Social Relations to Enhance Communication
Strategies

Mees A. Lindeman
12231207

Bachelor thesis
Credits: 18 EC

Bachelor *Kunstmatige Intelligentie*



University of Amsterdam
Faculty of Science
Science Park 900
1098 XH Amsterdam

Supervisor
dr. R. G. Alhama & dr. P. Le

Informatics Institute
Faculty of Science
University of Amsterdam
Science Park 900
1098 XH Amsterdam

Semester 1, 2023-2024

Abstract

This thesis explores the impact of structured input on the development of communication strategies in artificial agents, aiming to foster human-like language through compositional generalization. Utilizing graph-structured environments, the study investigates how agents evolve language within a setup that simulates social interactions and relations. The findings reveal that agents achieve high task-solving accuracy and exhibit a positive correlation between message entropy and accuracy, indicating a propensity for complex language strategies in larger, more challenging environments. However, this complexity does not necessarily align with the efficiency of human language, as evidenced by higher entropy levels diverging from the minimization observed in human language. While smaller game sizes prompt more compositional language structures, complexity in tasks does not invariably lead to increased compositional language, challenging the assumption of its necessity for effective communication. The study underscores the effectiveness of graph-structured inputs in promoting topographic similarity, suggesting potential avenues for research into more human-like language structures in multi-agent systems. Despite its contributions, the study notes limitations, including the chosen vocabulary size, which may have influenced the observed entropy. Overall this study adds valuable insights into the role of structured environments in the development of artificial languages, highlighting the complexities and challenges of mimicking human-like compositional generalization.

Contents

1	Introduction	3
2	Related Research	5
2.1	Language Emergence in Multi-Agent Systems	5
2.2	Graph-Based Approached Language Emergence	6
2.3	Emerging Language	7
2.3.1	Topographic similarity	8
2.3.2	Message Entropy	8
3	Methods	10
3.1	Dataset requirements	10
3.2	Data generation	11
3.3	Data splitting and Batching	13
3.4	Network	13
3.4.1	GATv2Conv	13
3.4.2	TransformerConv	14
3.5	Agents	14
3.6	Training	15
3.6.1	Hyperparamaters	15
3.7	Emerging Language	16
4	Results	17
4.1	Performance	17
4.2	Language emergence	18
4.2.1	Entropy	19
4.2.2	Topographic similarity	20
4.2.3	Emerging messages	22
5	Discussion	25
6	Conclusion	27

A	Tables	29
B	Figures	31

Chapter 1

Introduction

Human language’s unique capability to generate an infinite array of sentences from a finite vocabulary allows for expressing and understanding complex hierarchical and relational concepts, such as kinship relations and logical deduction chains (Hauser et. al, 2014) [1]. This form of linguistic adaptability refers to the ability to understand new combinations of words based on known elements. For example, if someone is familiar with the phrases ‘red circle’ and ‘blue square’, they can intuitively grasp the meaning of a new phrase like ‘red square’ [2]. This ability to construct larger linguistic expressions by combining simpler parts is known as compositionality (Baroni et. al, 2019) [3]. As interest raises in developing artificial agents evolving communication to better accomplish cooperative tasks, the question arises of how the emergent language can be bestowed with the same desirable compositionality property (Chaabouni et al., 2020) [4].

In the domain of emergent communication, artificial agents develop communication protocols from the ground up, often through referential games (Lazaridou & Baroni, 2020) [2]. By viewing communication as an interactive, goal-oriented challenge, some of the problems seen in supervised learning—such as the lack of samples and the tendency to remember superficial signals instead of uncovering the true factors of variation in the data—can be reduced (Baroni, 2020) [3]. However, compositional generalization, a concept deeply ingrained in human communication, remains a challenge for current multi-agent systems, even when they are trained with restricted symbolic data (Lazaridou et al., 2018) [5].

Human communication often revolves around social themes, with roughly two-thirds of conversational time typically devoted to what can broadly be termed as gossip [6]. This inclination towards social topics underlines a deeper cognitive ability: ability to represent complex concepts and the relationships between them in the form of a mental graph which has been identified as a major contributor to the generalization of knowledge and prolonged learning in humans [7]. Building on

this understanding, Słowik et. al (2020) [8] investigated the use of graph representations in place of traditional image-based inputs in referential games, attempting to use structural inductive biases to develop a more compositional language.

This study aims to further explore the the potential of graphs—where relationships are explicitly named and structured—for fostering more human-like language in multi-agent systems. By doing so, it seeks to mimic a web of social interactions and relations inherent in human communication, thereby guiding artificial agents to develop more human-like languages and communication strategies. By understanding the natural human propensity for gossip and our ability to mentally organize complex relationships, this study hypothesises that artificial agents can achieve a higher degree of compositional generalization when presented with a set up that mimics these interactions. Through this approach, the study aims to bridge the gap between the compositional flexibility of human language and the current limitations faced by multi-agent systems.

Chapter 2

Related Research

The study of emerging language, particularly through multi-agent communication games, is important for two related problems: understanding language evolution; and language acquisition—artificial intelligence’s (AI) ability to learn existing language [2]. This dual approach addresses two interconnected facets: understanding the evolutionary processes that might have shaped the properties of human language, such as compositionality—allowing for the creation of infinite expressions from a limited set of elements (Baroni, 2019) [3]—and refining AI technologies to better grasp and generate human-like language. The latter being an increasingly difficult task as it is believed that supervised learning alone is not going to provide a solution for this challenge [9].

2.1 Language Emergence in Multi-Agent Systems

Much research has been done in the exploration of language emergence within multi-agent systems. For example, Lazaridou et al. (2017) [10] have demonstrated that agents can develop a form of language through image-based referential games. A referential game consists of two agents with distinct roles: the speaker, with access to some form of input data and a target objective, and the listener, who has access to the same data but has no knowledge of the target objective. The speaker’s role is to convey information about the target to the listener through a descriptive message. The listener then receives this message and must identify the target amidst a set of distractors—other objects randomly selected from the same data distribution as the target object. Success in the game is measured by the listener’s ability to correctly identify the target, with rewards given solely for accurate recognition and not for the content of the messages exchanged (Lazaridou et al., 2018; Lazaridou & Baroni 2020) [5, 2]

To increase the level of natural language understanding and compositionality,

further strides in this area were made by Havrylov & Titov (2017) [9], comparing reinforcement learning with a differentiable relaxation method using the Gumbel-Softmax estimator. This trick enables the approximation of discrete categorical variables with differentiable continuous variables using a temperature-controlled softmax function, allowing for gradient-based optimization. This makes the system fully trainable via backpropagation. By doing so they discovered that the Gumbel-Softmax estimator converged faster and resulted in more effective communication protocols. Furthermore, enabling agents to use sequences of discrete tokens rather than single tokens revealed that their model developed language assigning distinct meanings to positions and sequences, suggesting hierarchical coding. For example, sequences with "5747" at the beginning signals the presence of an animal, and certain combinations like "5747 5747 7125" specifically refer to bears being present. This demonstrates properties of natural languages, such as compositionality (Havrylov & Titov, 2017) [9].

Other studies have explored the subject further. Ohmer et al. (2021) [11] focused on studying the perceptual processes related to how agents perceive and differentiate object features, such as color, size, and shape. These perceptual processes are crucial because they determine the way agents encode and communicate information about their environment, ultimately influencing the structure and properties of the emergent language. This approach led to the development of emergent languages that not only performed better in task-based communication games but also closely mirrored the conceptual structure of the inputs. Significantly, this work revealed that biases inherent in standard neural network models deeply affect language formation. A finding supported by Li & Bowling (2019) [12], who show emerging language is often the result of specific environmental pressures during training. These findings suggest subsequent research to delve into richer visual domains, promising to enhance the sophistication of communication in multi-agent systems. (Ohmer et al., 2021) [11]

2.2 Graph-Based Approached Language Emergence

Despite increasing the complexity of inputs through image-based methods, it has become clear that simply adding more information does not necessarily lead to improved outcomes in language compositionality or generalization. This was notably highlighted by Lazaridou et al. (2018) [5], who found that richer inputs don't always yield better linguistic results. Battaglia et al. (2018) [13] have posited that graph representations hold greater promise for fostering stronger compositional generalization in classification tasks.

Graph Neural Networks (GNNs) represent a significant advancement in the field of deep learning, particularly in processing and learning from graph data (Zhou

et. al, 2020) [14]. These networks are adept at managing data that embodies rich relational information among its elements. The core mechanism of GNNs involves message passing between the nodes of a graph, effectively capturing the dependencies within the graph structure. This enables them to reason about and learn from both structured data and non-structured data, by analyzing extracted structures like dependency trees or scene graphs (Zhou et. al, 2020) [14]. In the context of referential games, the use of GNNs to properly capture graph input aligns well with recent findings that simply increasing the quantity of inputs, such as images, does not necessarily lead to more sophisticated or compositional rich language emergence.

In their research on referential games using graph input, Słowik et al. (2020) [8] introduced a graph-based approach to emergent communication in multi-agent systems. They expanded traditional referential games by incorporating GNNs as a structural basis for input representation. This innovative method allowed agents to interact within a framework that reflects more complex and hierarchical data structures, akin to real-world scenarios. Their work demonstrated that using graph-based inputs, compared to bags-of-words or sequence representations, leads to the development of a more compositional language by the artificial agents (Słowik et al, 2020) [8]. This compositional language facilitated a systematic generalization to new combinations of known features, showcasing the potential of GNNs in enhancing the capability of multi-agents systems to process and communicate complex relational information. This highlights the potential graph-based input has for a more compositional language emerging. Improving on (Słowik et al, 2020) [8] this study also tries to incorporate human-like elements such as family relationships and characteristics.

2.3 Emerging Language

Throughout this study, the concept of compositionality has been mentioned several times. In linguistics, compositionality refers to human’s ability to make an infinite use of finite means. Formally a small set of elements-such as words-can be composed in limitless ways, such as into new sentences (Battaglia et. al, 2018) [13]. Research in the field of language emergence in multi-agent systems has focused on how compositional languages can emerge as a result of the interactions between agents. Agents often develop their own form of communication to achieve specific goals or complete tasks. The emergence of compositionality in these languages is a significant area of interest because it mirrors the fundamental aspects of human language, which is inherently compositional (Hauser et. al, 2014) [1]. An intuitive understanding of compositionality is shown in Figure 2.1.

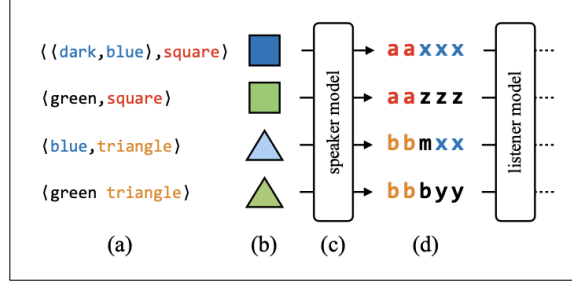


Figure 2.1: Representations arising from a communication game.

In this game, an object (b) is presented to a learned speaker model (c). The speaker model encodes the object as a discrete character sentence (d) to be interpreted by a listener model. The space of inputs has known compositional structure (a). The goal is to measure to what extent this structure is reflected in the structure of the learned codes. Image taken from Andreas (2019). [15]

2.3.1 Topographic similarity

There is no definitive quantitative measure of language compositionality. However, a number of studies in the field of language emergence use a quantitative measure of message structure referred to as topographic similarity (TopSim). Given a set of objects, their meanings and the associated signals, Brighton & Kirby (2006) [16] define topographic similarity to be the correlation of the distances between all the possible pairs of meanings and the corresponding pairs of signals. They have shown that mappings between states and signals show higher TopSim for compositional languages in comparison with holistic languages. This method of calculating compositionality was reintroduced in the field of language emergence by Lazaridou et. al (2018) [5] and has been widely adapted in numerous studies involving referential games.

Generally, to compute this measure two lists of numbers, corresponding to the input space and the message space, are calculated using a predefined method (i.e. cosine similarity). Given these two lists, the topographic similarity is defined as their negative Spearman ρ correlation. Intuitively, if similar objects share much of the message structure (e.g., common prefixes or suffixes), and dissimilar objects have little common structure in their respective messages, then the topographic similarity should be high, the highest possible value being 1.0 [5].

2.3.2 Message Entropy

Aside from topographic similarity the concept of entropy has been a topic of interest in language analysis. Entropy is a metric that quantifies the amount of uncertainty in a given state. In the context of language processing, entropy re-

duction defines the extent to which a word decreases the amount of uncertainty about what is being communicated [17]. Research by Karitonov et al. (2020) [18] investigates the complexity of languages that emerge when artificial agents are trained to communicate through a discrete channel.

Communicative efficiency, observed across various semantic domains, tends to minimize lexicon entropy while maintaining accuracy (Zaslavsky et al., 2019) [19]. For instance, most languages differentiate between grandmothers and grandfathers, but few distinguish between maternal and paternal grandmothers due to the marginal increase in accuracy versus the complexity it adds to the lexicon [19]. Kharitonov et al. (2021) [20] demonstrated that emergent languages in artificial agents are subject to entropy minimization pressures. These pressures aim to minimize the mutual information between the agent’s inputs and messages, leading to more robust representations against overfitting when driven by discrete channels (Kharitonov et al., 2021) [20]. Separately, Gibson et al. (2019) [21] provided evidence that similar efficiency pressures are at work in human language and other biological communication systems. This suggests a parallel between the entropy minimization observed in artificial agents and that in human language.

Chapter 3

Methods

First, a custom dataset is generated containing graphs that model family structures, adhering to specific rules involving age and gender. Subsequently, data is batched and processed through custom DataLoader classes. Two neural network agents, a Sender and a Receiver, are trained to encode and decode messages about relationships representing the graph input. Training effectiveness is evaluated on the Receivers ability of classifying a predefined target node. The emerging language is then analyzed by using topographic similarity and entropy as metrics.

3.1 Dataset requirements

The objects in the dataset were designed to serve as input for sender and receiver agents engaged in a referential game. The core objective of this game is to enhance the agents' ability to recognize and interpret human relationships, an important aspect in this study since it posited to mimic social interactions inherent in human communication.

To ensure a realistic representation, graphs are not just a collection of random names and ages; they are carefully constructed models that aim to reflect the nature of real-life families. The generation of a single graph is guided by two preset rules:

- **Spousal Age Difference:** Reflecting common marital patterns, the age difference between spouses is capped at five years. This constraint not only upholds the realism of the graph but also regulates the generational progression, ensuring that age descent per generation remains plausible.
- **Children's Age Gap:** To establish a realistic generational gap, children are generated to be 20 to 30 years younger than their youngest parent. This rule too contributes to upholding a controlled generational progression.

Graphs are generated like a family tree in a breadth-first manner, starting with an elder (root) generation with ages set in a 80 to 100-year range. From this starting point, the family tree grows iteratively. Children are added to parents following the aforementioned age rules, which are then given new spouses, continuing the process. Through each generation new family members are linked to their predecessors, forming a web of social relationships.

In addition to the previously set rules, the criterion for spousal relationships requires a binary gender system. Specifically, spouses are designated as male and female, a decision made to facilitate straightforward childbearing processes within the generation. This rule serves to streamline the family tree generation, allowing for a more focused analysis of family relationships without the complexities of a more diverse gender system.

3.2 Data generation

In the dataset generated for this study, the underlying graph structure is utilized to model family relationships and individual characteristics. This graph $G = (V, E)$ consists of vertices V and edges E , where each vertex represents an individual in the family tree, and the edges denote family relationships. The following features and attributes included in the dataset are designed to help the sender and receiver agents process and interpret family structures properly:

1. **Node Features (Vertices):** Each vertex $v \in V$ is associated with a set of node features $f(v)$. In this context every individual in the family tree, represented by a vertex, is characterized by two specific features:

- Age (denoted as $f(v_{age})$)
- Gender (denoted as $f(v_{gender})$)

These features not only help the agents to distinguish individuals but also play a significant role in the iterative process of generating graphs.

2. **Edge Attributes:** The edges in the graph, $e \in E$, represent possible family relationships and are defined by one of three attributes:

- 'married': defines a bidirectional relationship, signifying a marital bond between spouses. For spouses s_1 and s_2 , the edges are represented as (s_1, s_2) and (s_2, s_1) with the attribute $\text{attr}(e_{s_1 s_2}) = \text{'married'}$ and $\text{attr}(e_{s_2 s_1}) = \text{'married'}$, respectively.
- 'child-of': defines an unidirectional relationship from a child to their parents. Formally, an edge e_{cp} from child c to parent p is represented as

(c, p) with the attribute $\text{attr}(e_{cp}) = \text{'child-of'}$. Forms the lineage within the family structure together with 'gave-birth-to'.

- 'gave-birth-to': defines an unidirectional relationship from a parent to their child. Formally, an edge e_{pc} from parent p to child c is represented as (p, c) with the attribute $\text{attr}(e_{pc}) = \text{'gave-birth-to'}$.

Incorporating node features and edge attributes into the graph G provides a framework for capturing relationships. The inclusion of these specific characteristics is important to this study, as they lay the biases for the agents to effectively navigate the relationships inherent in family structures.

When generating the dataset, a final parameter g is passed to indicate the number of generations. This provides a flexible way of controlling the size of the graphs. Figure 3.3 provides two examples of generated graphs: one with two generations and another with three. The two-generation graph is simpler, great for easy understanding and showcasing the essence of iteration. The three-generation version, however, embodies a more complex family network, showing how graphs grow and evolve over each iteration.

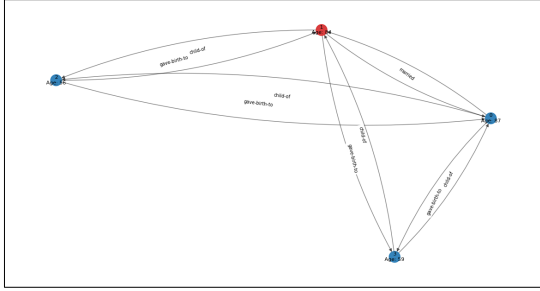


Figure 3.1: 2 generations

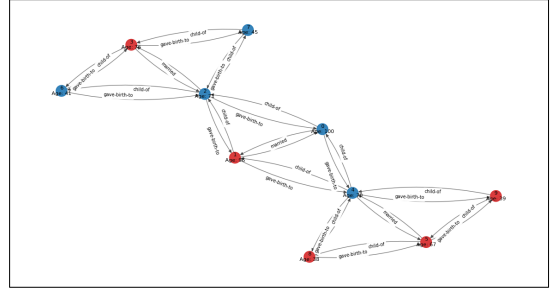


Figure 3.2: 3 generations

Figure 3.3: Example of two family graphs

For these graphs, the amount children c is set to 2, leading to different graph sizes based on the number of generations g : $|V| = 4, 10$, and 22 for $g = 2, 3$, and 4 respectively. Nodes are indexed starting at 0, formally v_i for the i th node, allowing for simple reference. Node features, $f(v_i)$, are represented as follows: for categorical features, one-hot encoding is utilized resulting in $f(v_{gender})$ being a vector where only one element is 1 and the rest are 0; for numerical features, $f(v_i)$ is a vector representing continuous values associated with $f(v_{age})$.

3.3 Data splitting and Batching

Sotrin the graphs is done using the PyTorch Geometric (Fey & Lenssen, 2019) 'Data' class, holding a range of attributes by default and optionally more if required. For this research a target node is randomly selected in each graph and a sequence is generated describing the graph structure, both are then passed as additional attributes. A custom DataLoader was made for this study to return a batch of graphs as auxiliary input for the agents and a collection of randomly selected target nodes as labels. The receiver input is left empty as the batch is processed in the agents individually. The sender input is given by a sequence describing the edges $e \in E$ because it is needed for computing topographic similarity, which will be explained in later sections.

Three datasets, denoted as D_1 , D_2 , and D_3 , each containing 3200 graphs were created using the aforementioned method of data generation. The size of each dataset is expressed as $|D_1| = |D_2| = |D_3| = 3200$. In the datasets D_1 , D_2 , and D_3 , the graphs $G = (V, E)$ contain $|V(D_1)| = 4$, $|V(D_2)| = 10$, and $|V(D_3)| = 22$ nodes respectively. These node counts are achieved by specifying the number of generations g as 2 for D_1 , 3 for D_2 , and 4 for D_3 , while setting the number of children c to 2 for each parent couple in all datasets. An allocation of 10% test data, 10% validation data, and the remaining 80% as train data for each dataset was used during training leading to 2560 train samples, 320 validation samples and 320 test samples whilst batch size was set to 32.

3.4 Network

Two types of encoders were employed in this study, using distinct layers integrated in PyTorch Geometric (Fey & Lenssen, 2019) [22]: 'GATv2Conv' and 'Transformer-Conv'. This choice was driven by their use of multi-head attention mechanisms, which are adept at capturing node relationships in graphs, likely enhancing the model's capability to learn nuanced representations of graph data.

3.4.1 GATv2Conv

Graph Attention Networks (GATs) are a type of GNN that update node representations by attending to their neighbors using their own representations as benchmark. This approach allows nodes to compute a weighted average of their neighbors' features, selecting the most relevant neighbors (Veličković et al., 2018) [23]. The dynamic attention mechanism—an improvement on regular GAT layers—makes 'GATv2Conv' more robust, particularly in situations where the relationships between nodes vary across different parts of the graph (Brody et. al, 2021) [24].

The GAT encoder is defined $f^G(g, \theta_f^G)$ and utilises two 'GATv2Conv' layers, leaky ReLU is applied as activation function between the two layers. The first layer has a number of attention heads which can be set as a hyperparameter (see 3.6 Training), the second layer uses one attention head for output purposes.

3.4.2 TransformerConv

The 'TransformerConv' layer integrates the principles of the Transformer model, renowned in Natural Language Processing (NLP), into graph neural networks (Shi et al., 2021) [25]. This multi-head attention mechanism used in NLP is adapted for graph data by computing multi-head attention for each edge in the graph, taking into account node features and edge attributes. By leveraging multi-head attention, TransformerConv can focus on different aspects of node information and their connections, leading to more accurate and refined node classification (Shi et al., 2021) [25]. The Transform encoder is defined $f^T(g, \theta_f^T)$ and utilises two 'TransformerConv' layers. Similar to the GAT encoder $f^G(g, \theta_f^G)$ leaky ReLU is applied as activation function. Additionally, the first layer shares the number of attention heads with its GAT counterpart, set by the same hyperparameter (see 3.8 Training), while the second layer uses a single attention head for its output.

3.5 Agents

Two neural network modules were used as agents in the referential game. The Sender module π^S was created as the communicative agent responsible for encoding and transmitting the description of a target node. The Sender encodes a graph g from a dataset D_i , where $g \in D_i$, into a dense representation u using encoders $f^T(g, \theta_f^{T,S})$ and $f^G(g, \theta_f^{G,S})$. Hereby utilizing the attention mechanisms to focus on relevant aspects of the input data. Each graph contains a predefined target node, the graph embedding u is indexed at the specific node and gets processed by a linear transformation decoder $g^S(u, \theta_g^S)$ layered with a *GRU* cell, resulting in an encoded message m describing the target. The Receiver module π^R serves as the interpretive agent, designed to decode and understand the message m received from the Sender. It shares the encoder architecture $f^T(g, \theta_f^{T,R})$ and $f^G(g, \theta_f^{G,R})$ with the Sender but has independent network weights. Upon receiving the message m , the module transforms it into an embedding m' through a linear transformation decoder $g^R(m, \theta_g^R)$, also layered with a *GRU* cell. This embedding is then compared with the processed input data u' computed by its encoders to ascertain its relevance and meaning. The outcome of this comparison is expressed as a log softmax distribution over the range of nodes, representing the agent's interpretation of the received message and classification of the correct node.

3.6 Training

All weights of the sender and receiver agents, $\theta = \{\theta_f^{G,S}, \theta_f^{T,S}, \theta_g^S, \theta_f^{G,R}, \theta_f^{T,R}, \theta_g^R\}$, are jointly optimized while playing the game. No weights are shared between the two agents and the only supervision used is communicative success, i.e. whether the listener identified the correct target node. Similar to Słowik et al. (2020) [8], a Gumbel-Softmax estimator was used during training to make the message discrete—allowing for better model optimization. It was also used by Havrylov & Titov (2017) [9], who showed using Gumbel-Softmax results in more efficient learning strategy. The agents minimize a Negative Log-Likelihood (NLL) loss function during training:

$$\text{Total NLL Loss} = \frac{1}{N} \sum_{i=1}^N -\log(\hat{y}_i[y_i]) \quad (3.1)$$

where $\hat{y}_i$ denotes the predicted probability distribution for the i -th data point, and y_i is the true label. Accuracy is calculated as the fraction of predictions where the predicted label matches the true label, as follows:

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N 1(\text{argmax}(\hat{y}_i) = y_i) \quad (3.2)$$

Here, $1(\cdot)$ is the indicator function, which is 1 if the condition is true and 0 otherwise.

3.6.1 Hyperparameters

During training, a range of hyperparameters were manually tuned and tested, using accuracy as the selection criterion. They are listed below, bold indicates the parameters that are reported in the results section:

- Embedding Size: **10**, 20, 40, 80
- Attention Heads: 2, **4**, 8
- Vocabulary Size: 10, 25, 50, **100**
- Max Length: **3**, **4**, **5**, **10**, 25
- Hidden Size: 10, **20**, 50
- Learning Rate: **0.01**, 0.001
- Sender-Receiver cell: **GRU**, LSTM, RNN

- Gumbel-Softmax temperature: **1.0**

These parameters were derived from the default settings of the EGG library (Kharitonov et al., 2021) [20], combined with parameters from the research conducted by Słowik, et al. (2020) [8]. Despite their research being of a different nature, they offer valuable insights into the processing of graph structures in referential games and served as a suitable starting point.

3.7 Emerging Language

Calculating topographic similarity requires an input space, to do this a method was devised to simulate a consistent graph structure. A sequence is generated based on a set of predefined rules and a vocabulary that translate the graph’s topological features into a numerical sequence.

First each node in the graph $G = (V, E)$ is identified by its features $f(v)$ for $v \in V$ and edge attributes $e \in E$. These characteristics are encoded into numerical values through the following vocabulary set *Vocab*, defined as:

$$Vocab = \{(' : 1, ') : 2, 'male' : 3, 'female' : 4, 'married-to' : 5, 'child-of' : 6, 'gave-birth-to' : 7\}$$

Starting from the target node, a tree structure is build. The branching process prioritizes certain edge attributes e (e.g., 'married-to') and features $f(v)$ (e.g., female over male), ensuring a consistent traversal of the graph. The tree is then converted into a sequence, represented by the numerical values $i \in Vocab$. Brackets are introduced in the sequence to denote extended branches, providing clarity in the hierarchical structure of the tree. Because different graphs and root nodes result in sequences of varying lengths, the sequences are standardized by padding shorter ones with dummy value '0'. The sequences generated from the graphs provide a consistent representation of its topology. To compute topographic similarity, the 'edit' metric is employed in both input and message spaces as they are sequence-based. Topographic similarity is quantified using the Spearman ρ rank correlation between the two spaces.

Similarly to Kharitonov et al. (2020) [18], resulting entropy H of messages m is explored for multiple game sizes. Calculating the entropy $H(m)$ of the distribution of discrete messages m is straightforward: message frequencies are converted to probabilities and then negative products of each probability are summed with its base-2 logarithm, measuring the collection’s uncertainty or diversity. Minimal entropy is set to $H_{min} = \log_2(V)$ where V denotes the amount of vertices $v \in V$, i.e. game size.

Chapter 4

Results

Multiple experiments were conducted, setting varying game sizes—interpreted as the number of distractors—and maximum message lengths to determine their influence on the agents’ performance. Representations of game sizes are found in B. The agents were trained over 300 epochs, and their performance was evaluated based on three main metrics: accuracy, message entropy, and TopSim. Accuracy is solely measured by the success of the Receiver agent π^R to correctly identify the target node in the graph, message entropy showcases the variability and unpredictability of the agents’ language, and TopSim assessed the degree to which the emerging language shows signs of compositionality.

The results are presented in a comparative manner to demonstrate the impact of game complexity and message length on the agents’ performance. The analysis provides insights into the balance between accuracy and the agents’ emergent language, as well as the conditions that promote compositional language development. In the following sections, detailed findings from the experiments are presented.

4.1 Performance

The results obtained for training and testing accuracy are presented in Figure 4.1. Agents demonstrated higher overall accuracy in games with fewer distractors (smaller game sizes). This suggests that reduced complexity in the environment allows agents to more easily identify and describe the correct relationships. Across all game sizes and message lengths, agents showed an improvement in accuracy over epochs. This improvement plateaued after a certain point, indicating a stabilization in learning. Notably, in smaller game sizes, agents reached this plateau faster, which could be attributed to less environmental complexity or overcomplication within the model. However, accuracy tends to decline after too many epochs in smaller game sizes, which could potentially be attributed to the model

exhausting the learning potential of the limited diversity in the data, leading to a slight drop in performance. Finally, agents have a slight tendency to over fit on larger game sizes, as is shown by the divergence between test and train accuracy. Nevertheless the convergence patterns between training and testing accuracy were mostly consistent, suggesting that the agents were able to generalize their strategies to unseen data, thus avoiding significant overfitting to the training set.

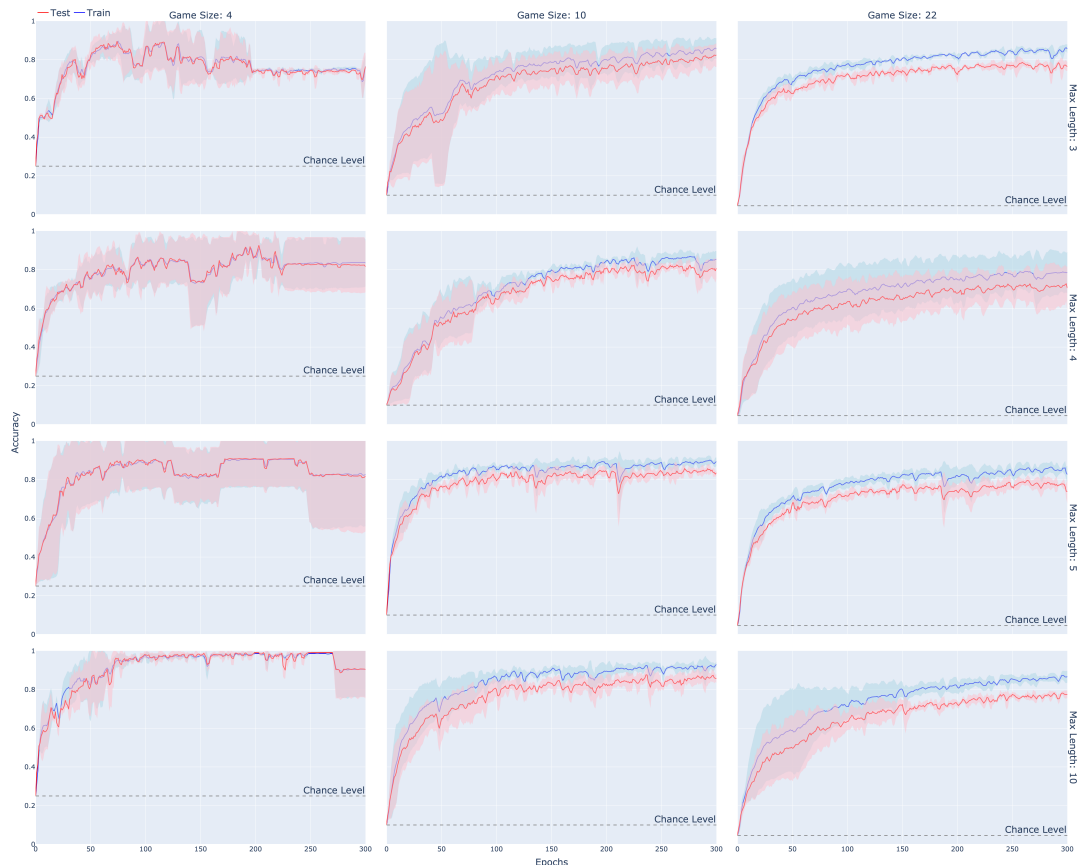


Figure 4.1: Test and training accuracy across varying message lengths and game sizes.

Mean and standard deviation are reported across three random seeds.

4.2 Language emergence

The results obtained in this study offer insights in artificial agents' ability to develop language under constraints that are designed to mimic human language. The interplay between the variability of their message construction (entropy), and

the structural alignment of their language with semantic meanings (TopSim) is examined.

4.2.1 Entropy

As shown in Figure 4.2 there is an increase in message entropy $H(m)$ across all message lengths when game size increases, indicating greater variability or complexity when the game involves more information.

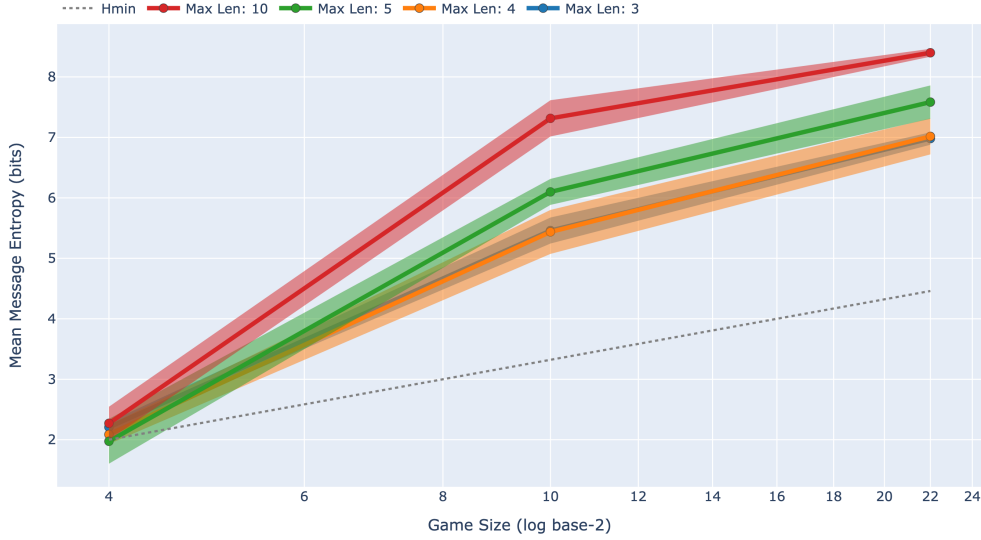


Figure 4.2: Message entropy $H(m)$ compared to game size for epoch 300. Mean entropy and standard error of the mean are reported across three random seeds.

The figure shows that as the game size increases, the mean message entropy $H(m)$ also increases, and it does so at a rate faster than the minimal entropy H_{min} . Possibly suggesting that as the complexity of the task increases, emergent language become more complex or uncertain. Larger message lengths are associated with higher overall entropy. This could mean that when the agents have the flexibility to use longer messages, they tend to use this flexibility to encode more information. It is worth observing that the message entropy $H(m)$ for message length 5 is slightly below H_{min} . The variability in accuracy for this message length, as depicted in Figure 4.3, may suggest that agents struggle to communicate sufficient information for task success, which correlates with the observed entropy. The subsequent section will specifically investigate the relationship between accuracy and entropy. The precise values for message entropy, accuracy, and additional metrics are summarised in Table A.1 of Appendix A.

An increase in accuracy was generally accompanied by an increase in message entropy, as can be seen from Figure 4.3. This positive correlation suggests that agents develop more complex and less predictable language to succeed at describing the target node.

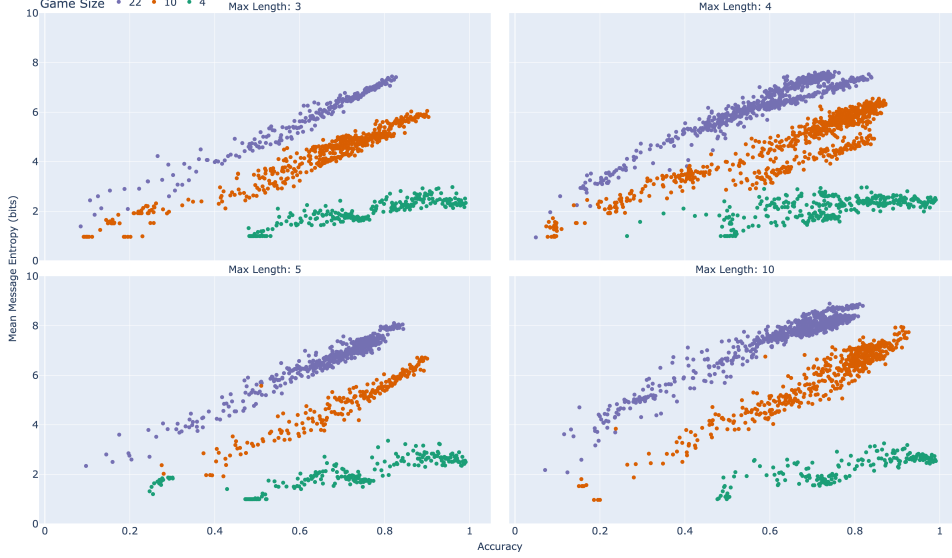


Figure 4.3: Message entropy $H(m)$ compared to accuracy. Data points represent averaged entropy across three random seeds.

Larger game sizes, which presented more distractors, are associated with higher entropy, indicating that more complex social environments necessitated a broader and more diverse use of language to maintain accuracy. Overall it is observed that message entropy $H(m)$ increases with game size, reflecting greater variability or complexity in emergent language as the task becomes more complex and correlates positively with accuracy.

4.2.2 Topographic similarity

Unlike accuracy shown in Figure 4.1, TopSim showed no constant increase across epochs, illustrated in Figure 4.4. This suggests that the compositional structure of the agents' language was subject to change as learning progressed. In a larger game size, high accuracy was achieved without a corresponding increase in TopSim, indicating that successful communication does not always necessitate a compositional language. The complexity introduced by larger game sizes and the flexibility offered by longer message lengths appeared to influence TopSim. However, the re-

relationship was not as straightforward as with accuracy and entropy, indicating that compositional language development is affected by multiple factors.

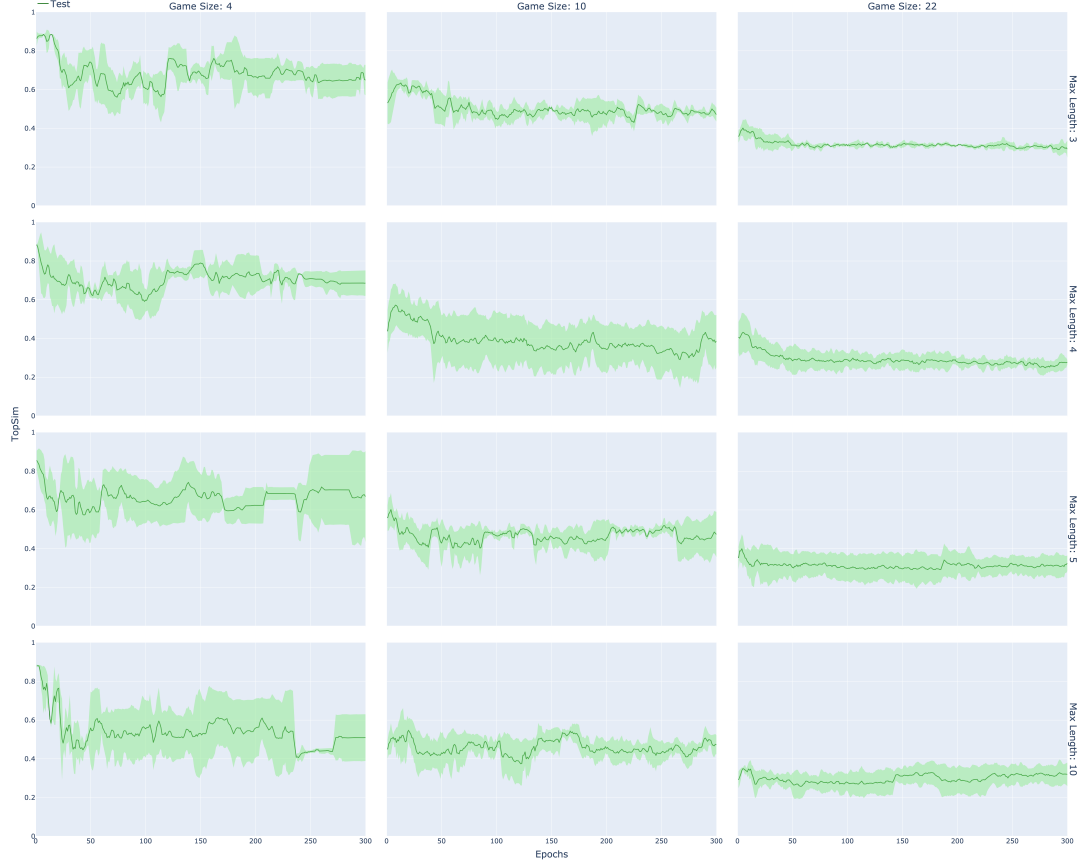


Figure 4.4: TopSim across varying message lengths and game sizes.
Mean and standard deviation are reported across three random seeds.

In Figure 4.5 an average of TopSim across three random seeds is computed, it can be observed that TopSim is high for a smaller game size when compared to a larger one. Additionally, variance decreases as game size increases, underlining the results in Figure 4.4. The precise values for average TopSim on the final epoch are found in Table 1 of Appendix A.

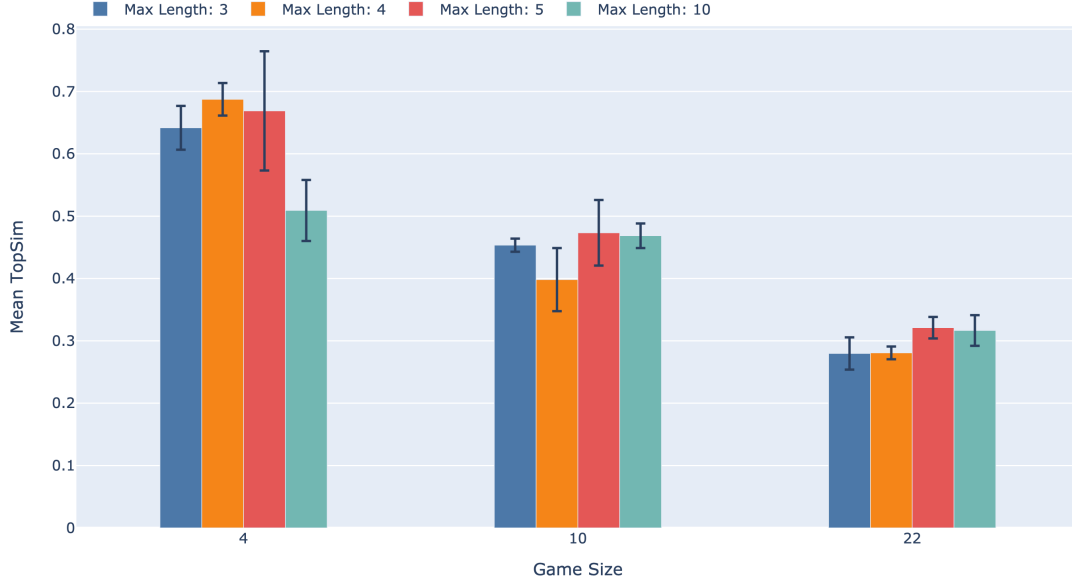


Figure 4.5: TopSim across varying message lengths and game sizes for epoch 300. Mean and standard deviation are reported across three random seeds.

Interestingly, the TopSim observed in this study exceeds the levels reported by Słowik et al. (2017) [8], with our results showing a marked increase over their documented TopSim of approximately 0.14 for 19 distractors. It is important to note that the experimental setups between the two studies differ. Yet this discrepancy highlights the complexity and type of interactions represented by graphs play a crucial role in the development of compositional language.

4.2.3 Emerging messages

Throughout this study, a slight negative correlation between message entropy $H(m)$ and TopSim has been observed. To further assess this pattern, Figure 4.2.3 compares the two metrics. There seems to be a noticeable trend where as TopSim increases, the entropy decreases. This is visualised as a diagonal gradient from the bottom right to the top left, specifically detailed in the top left subplot (message length 3) but it can also be seen in Table A.1. This suggests that when the language used by agents is more similar to the input (high TopSim), the messages tend to be more predictable (low $H(m)$), indicating a more structured communication system.

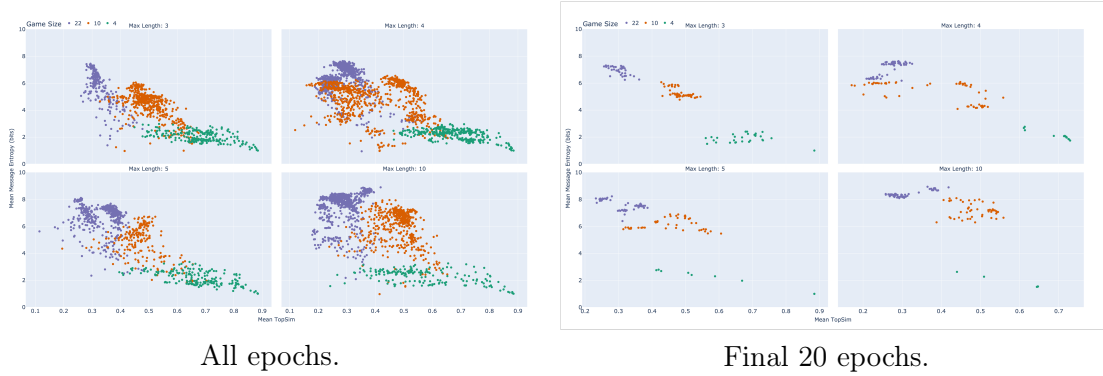


Figure 4.6: Message entropy $H(m)$ compared to TopSim.
Data points are averaged across three random seeds

When focusing on the last 20 epochs, where accuracy is generally higher, data points cluster at different $H(m)$ and TopSim regions depending on game size. For larger game size the agents may be using a more complex or abstract set of messages to represent the game state, possibly due to the necessity to convey more information due to the amount of distractors. For an intermediate game size, the agents seem to strike a balance between language structure (TopSim) and variability (entropy). This might reflect a phase where the language is sufficiently developed to capture the necessary details of the game without requiring the high variability necessary for larger game sizes.

The middle-range values could represent an efficient use of the language where it is neither too predictable nor too random. A small game size with low entropy and high TopSim suggests that the agents have developed a very structured language that closely mirrors the input. This could be due to the smaller game space requiring fewer distinct messages to communicate effectively. The low entropy indicates predictability and a limited set of messages, which might be all that is necessary to achieve high accuracy in a smaller game space. This is further illustrated in Figure 4.7, showing total message frequency for the top 10 messages in the last 50 epochs, where it can be observed that unique messages appear more frequent overall for a smaller game size.

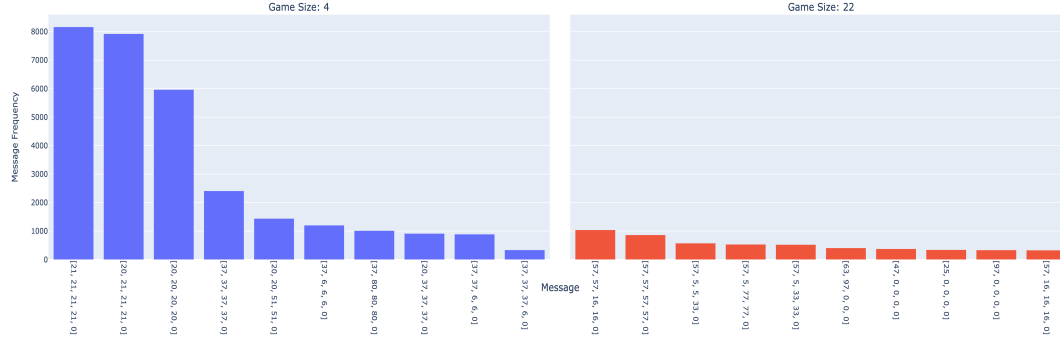


Figure 4.7: Unique frequencies for the last 50 epochs.
Explicit random seed with message length 4.

As illustrated in Figure 4.7, agents use less unique messages for a smaller game size as well as developing messages where symbols are repeated. In contrast, a larger game size utilises more varying messages, underlined by the difference in frequencies for the top 10 messages in Figure 4.7. This reflects a trade-off between the need for a rich language to describe a complex environment and the efficiency of communication in a simpler environment.

Chapter 5

Discussion

First and foremost, this study has shown that when provided with graphs as input, artificial agents develop messages that result in optimal task-solving accuracy. These results reflect those of Słowik et al. (2017) [8] who also found that the structural bias of graphs result in high accuracy. The reported message entropy $H(m)$, particularly for higher a game size, pointed to high variability in language strategies. This variability implies that agents may explore different linguistic pathways to success in more complex games, this was also illustrated by the message frequencies reported. The positive correlation between message entropy $H(m)$ and accuracy suggests agents did not converge on a simplistic or repetitive language to succeed at the task.

It is worth noting that while the emergent languages appear to be more varied for larger game sizes and longer message lengths, this does not mean they are less effective. It could be that the increased variety is necessary for successful communication in these more complex tasks. The results, however, also prompt a consideration of the efficiency of communication. While higher entropy indicates more complex language use, it also raises the question whether agents are using their language capacity optimally or if some of the variability is redundant. This could be attributed to the vocabulary size set for this study, which is quite large with 100. Finding a steeper increase compared to the minimal entropy H_{min} was unexpected and suggests that the emerging language does not follow sings of human-like language. As this outcome is in contrast to that of Kharitonov et. al (2020) [18], who showed human-like language has the tendency to minimize entropy.

Furthermore, the study found that topographic similarity is higher in smaller game sizes compared to larger ones. This suggests that in less complex environments, the compositional structure of the agents' language is more pronounced. Agents show rapid learning during initial epochs by developing highly composi-

tional language, thus increasing accuracy. However at a certain point they learn that there is no need for highly compositional language to complete the task, they develop a shortcut of sorts and possibly develop messages specific to the target node. In contrast, larger game sizes, which introduce more complexity, do not lead to a corresponding increase in topographic similarity, indicating that a compositional language structure is not always essential for effective communication in more complex settings. The higher topographic similarity observed in this study compared to previous findings (Słowik et al., 2017) [8] underlines the impact of structured inputs, such as graphs, on the development of compositional language. This highlights the potential to further explore structured environments to foster more human-like language structures in multi-agent communication systems.

Chapter 6

Conclusion

This study set out to gain a better understanding of the influence of structured input to develop communication strategies closer to human-like language in multi-agent systems. By simulating a web of social interactions and relations akin to human communication, this study sought to guide artificial agents towards developing communication strategies and languages that resemble human language, particularly in terms of compositional generalization.

The study demonstrated that artificial agents gain high task-solving accuracy and high topographic similarity when interacting within graph-structured environments, emphasising the effectiveness of structural biases inherent in graphs. Interestingly, a slight negative correlation was observed between message entropy and topographic similarity. This suggests that more complex communications tend to deviate from directly mirroring environmental cues, moving towards more abstract forms of expression. Consequently, this complexity can lead to a communication system that is less predictable in its direct mapping to inputs, despite being rich and varied.

Contrary to expectations and the minimization of entropy observed in human-like languages as reported by Kharitonov et al. (2020) [18], the emergent languages in this study exhibited high entropy levels, suggesting a departure from human linguistic tendencies. This divergence indicates that the language strategies developed by agents in this study do not necessarily align with human linguistic tendencies. Moreover, the study reveals that while smaller game sizes favor the development of a more compositional language structure, as evidenced by higher topographic similarity, this is not a prerequisite for effective communication in more complex tasks. This finding challenges the assumption that highly compositional language correlates to the agents' ability to increase accuracy at the task.

A shortcoming in this study is the initial emphasis on achieving high accuracy, which resulted in selecting a large vocabulary size and potentially led to

an overly intricate model, as evidenced by diminished accuracy in smaller game sizes. Future research should explore the implications of a reduced vocabulary size, hypothesizing that constraining agents to fewer symbols might enhance compositional generalization. The choice of vocabulary size was driven by the goal of maximizing accuracy, yet this approach inadvertently led to increased entropy in the emergent language, deviating from the anticipated outcome of promoting more human-like linguistic patterns. In spite of its limitations, this study adds to the understanding of language emergence through graph-based input. It is demonstrated that structurally biased inputs, when combined with scenarios mimicking human-like conditions, lead to a high degree of topographic similarity.

Acknowledgements

I would like to offer my sincere gratitude to my supervisors, dr. Raquel Garrido Alhama and dr. Phong Le for Their helpful feedback and involvement during this project.

Appendix A

Tables

Mode	Acc	TopSim	Entropy	Game Size	Max Length
train	0.77	-	-	4	3
test	0.78	0.64	2.21	4	3
train	0.85	-	-	10	3
test	0.84	0.45	5.46	10	3
train	0.84	-	-	22	3
test	0.76	0.28	6.98	22	3
train	0.83	-	-	4	4
test	0.8	0.69	2.09	4	4
train	0.85	-	-	10	4
test	0.83	0.4	5.44	10	4
train	0.78	-	-	22	4
test	0.65	0.28	7.02	22	4
train	0.83	-	-	4	5
test	0.81	0.67	1.97	4	5
train	0.89	-	-	10	5
test	0.84	0.47	6.1	10	5
train	0.83	-	-	22	5
test	0.77	0.32	7.58	22	5
train	0.9	-	-	4	10
test	0.9	0.51	2.27	4	10
train	0.94	-	-	10	10
test	0.87	0.47	7.31	10	10
train	0.87	-	-	22	10
test	0.78	0.32	8.4	22	10

Table A.1: Final metrics (epoch 300).
Averaged across three random seeds.

Appendix B

Figures

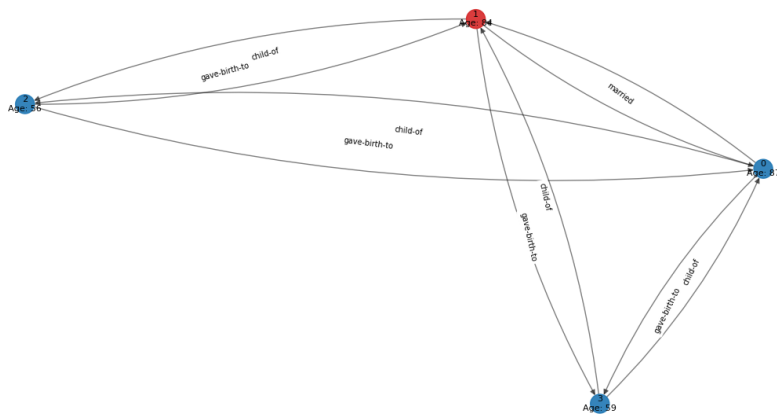


Figure B.1: Game size 4
2 generations

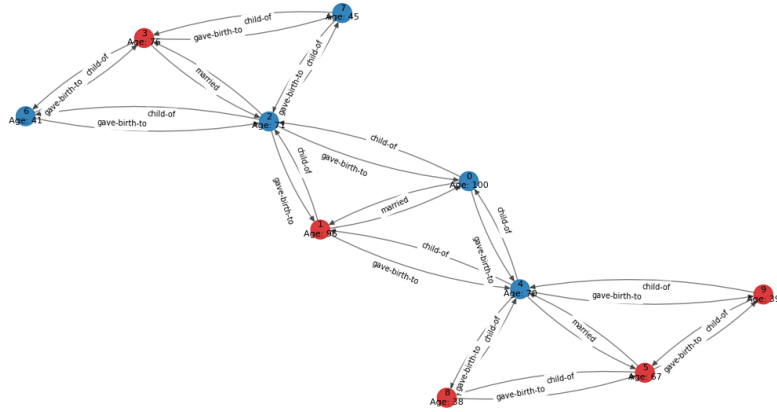


Figure B.2: Game size 10
3 generations

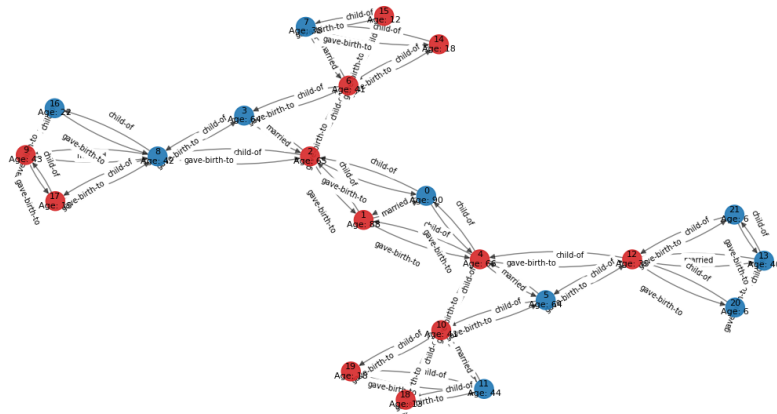


Figure B.3: Game size 10
4 generations

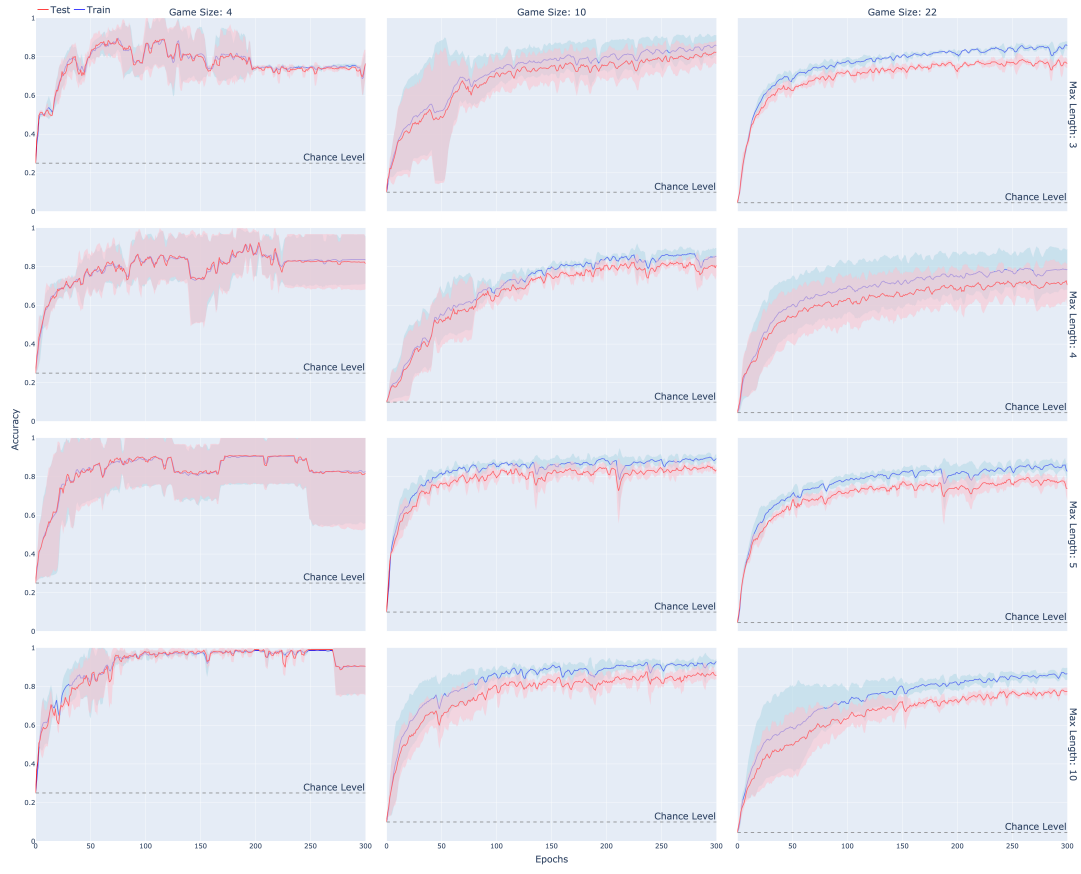


Figure B.4: Test and training accuracy across varying message lengths and game sizes.

Mean and standard deviation are reported across three random seeds.

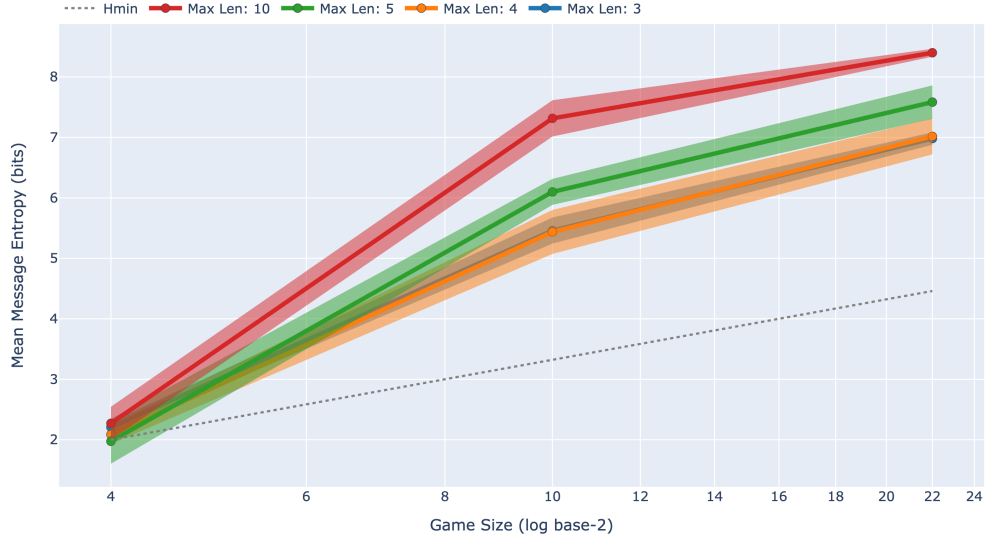


Figure B.5: Message entropy $H(m)$ compared to game size for epoch 300. Mean entropy and standard error of the mean are reported across three random seeds.

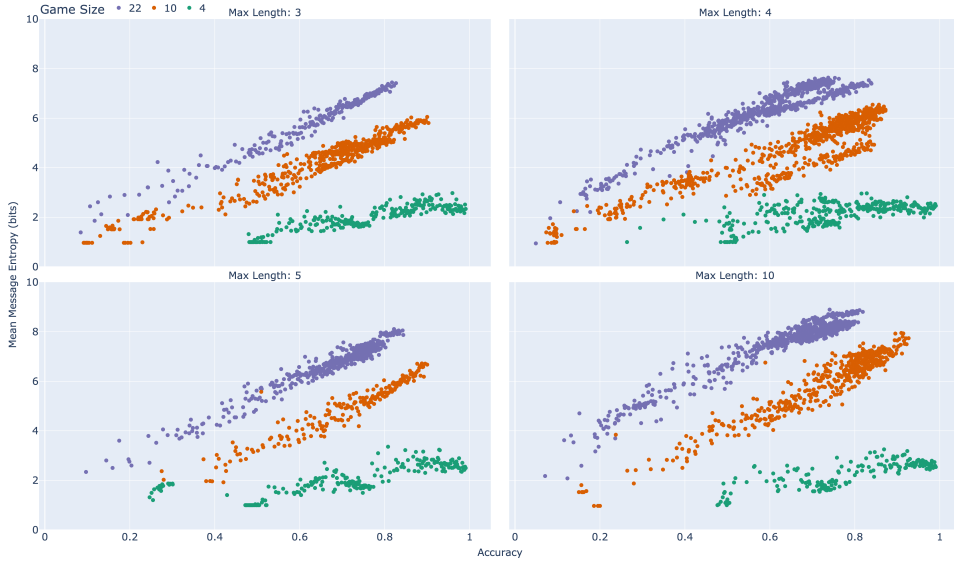


Figure B.6: Message entropy $H(m)$ compared to accuracy. Data points represent averaged entropy across three random seeds.

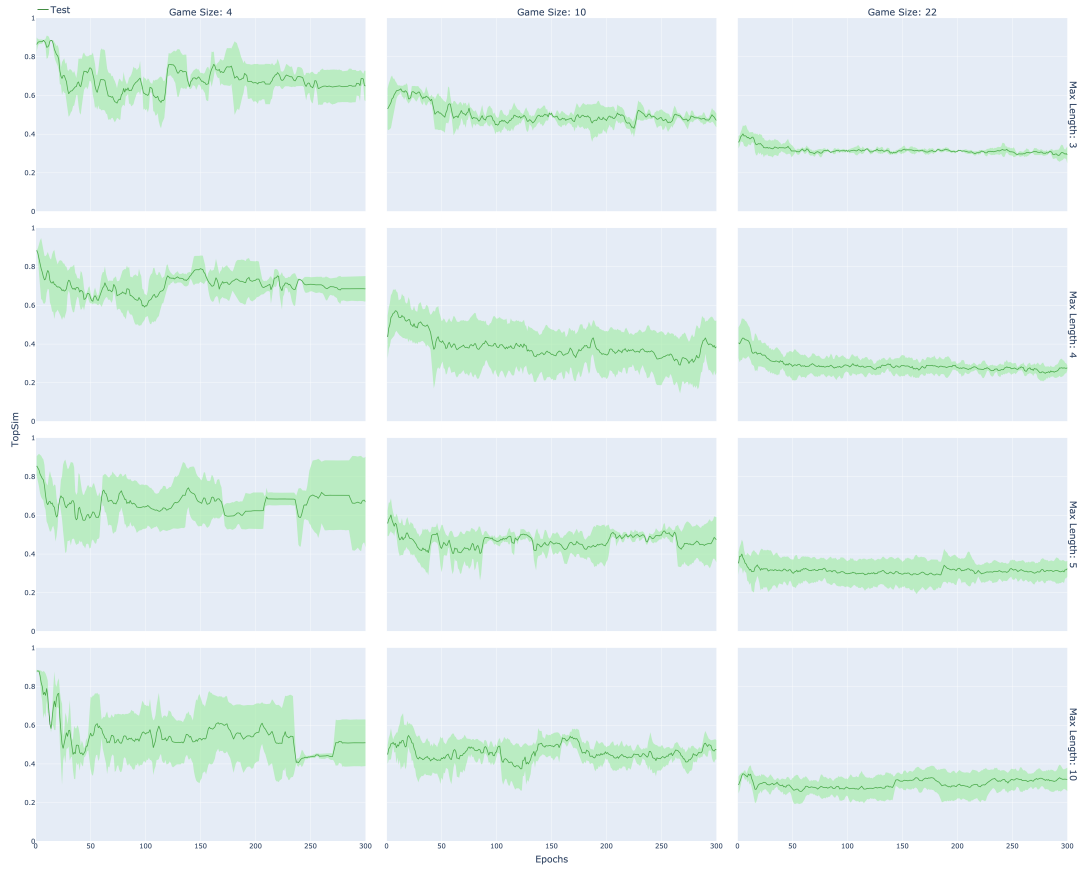


Figure B.7: TopSim across varying message lengths and game sizes.
Mean and standard deviation are reported across three random seeds.

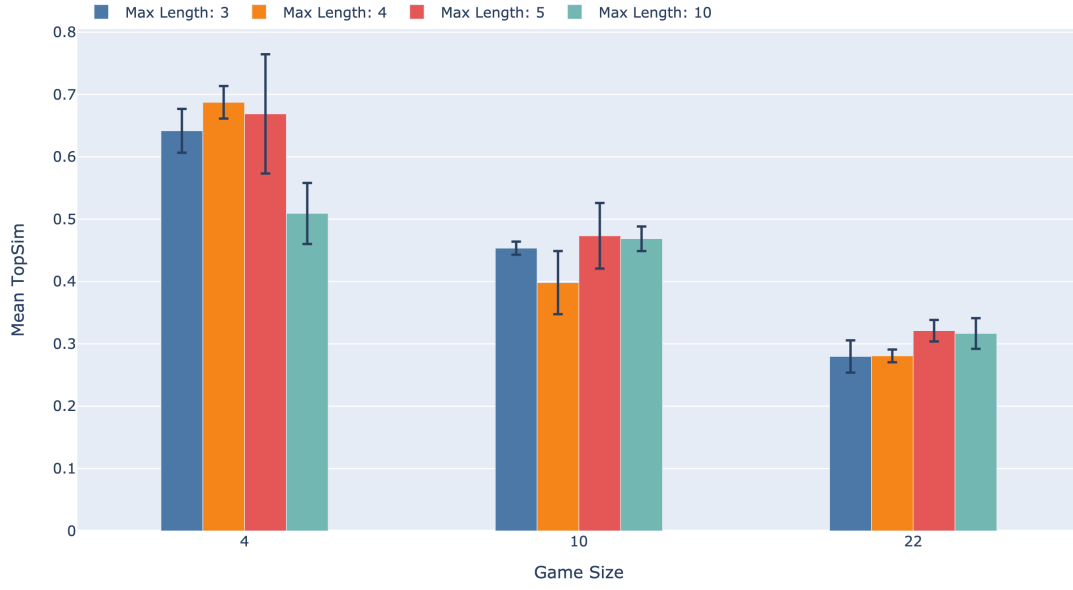


Figure B.8: TopSim across varying message lengths and game sizes for epoch 300. Mean and standard deviation are reported across three random seeds.

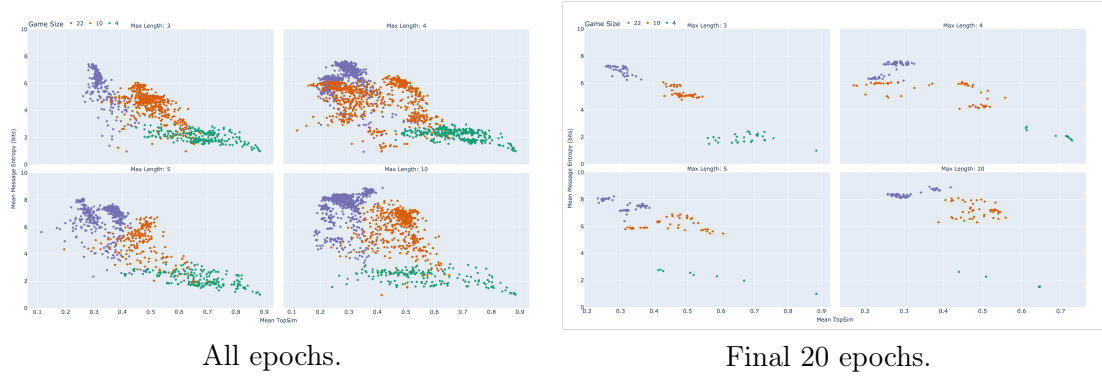


Figure B.9: Message entropy $H(m)$ compared to TopSim. Data points are averaged across three random seeds

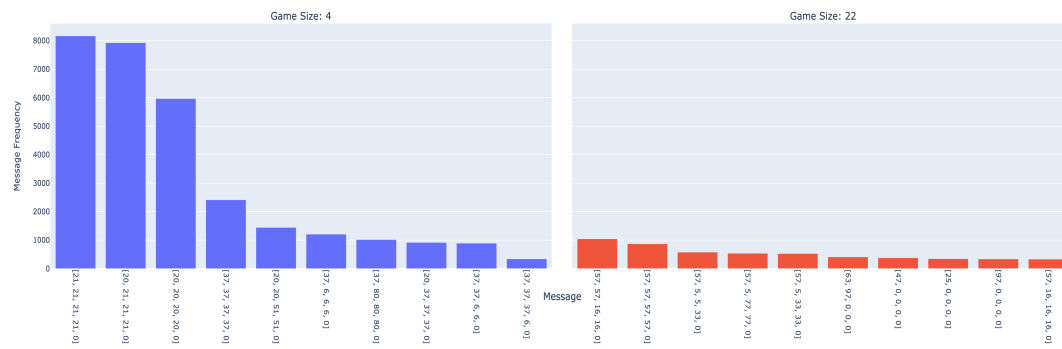


Figure B.10: Message frequencies for the last 50 epochs.
Explicit random seed with message length 4.

Bibliography

- [1] Marc D Hauser, Charles Yang, Robert C Berwick, Ian Tattersall, Michael J Ryan, Jeffrey Watumull, Noam Chomsky, and Richard C Lewontin. The mystery of language evolution. *Frontiers in psychology*, 5:401, 2014.
- [2] Angeliki Lazaridou and Marco Baroni. Emergent multi-agent communication in the deep learning era. *CoRR*, abs/2006.02419, 2020.
- [3] Marco Baroni. Linguistic generalization and compositionality in modern artificial neural networks. *Philosophical Transactions of the Royal Society B*, 375(1791):20190307, 2019.
- [4] Rahma Chaabouni, Eugene Kharitonov, Diane Bouchacourt, Emmanuel Dupoux, and Marco Baroni. Compositionality and generalization in emergent languages. *arXiv preprint arXiv:2004.09124*, 2020.
- [5] Angeliki Lazaridou, Karl Moritz Hermann, Karl Tuyls, and Stephen Clark. Emergence of linguistic communication from referential games with symbolic and pixel input. *CoRR*, abs/1804.03984, 2018.
- [6] Robin IM Dunbar. Gossip in evolutionary perspective. *Review of general psychology*, 8(2):100–110, 2004.
- [7] Jacob LS Bellmund, Peter Gärdenfors, Edvard I Moser, and Christian F Doeller. Navigating cognition: Spatial codes for human thinking. *Science*, 362(6415):eaat6766, 2018.
- [8] Agnieszka Słowik, Abhinav Gupta, William L. Hamilton, Mateja Jamnik, Sean B. Holden, and Christopher J. Pal. Exploring structural inductive biases in emergent communication. *CoRR*, abs/2002.01335, 2020.
- [9] Serhii Havrylov and Ivan Titov. Emergence of language with multi-agent games: Learning to communicate with sequences of symbols. *CoRR*, abs/1705.11192, 2017.

- [10] Angeliki Lazaridou, Alexander Peysakhovich, and Marco Baroni. Multi-agent cooperation and the emergence of (natural) language. *CoRR*, abs/1612.07182, 2017.
- [11] Xenia Ohmer, Michael Marino, Michael Franke, and Peter Köning. Why and how to study the impact of perception on language emergence in artificial agents. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 43, 2021.
- [12] Fushan Li and Michael Bowling. Ease-of-teaching and language structure from emergent communication. *Advances in neural information processing systems*, 32, 2019.
- [13] Peter W. Battaglia, Jessica B. Hamrick, Victor Bapst, Alvaro Sanchez-Gonzalez, Vinícius Flores Zambaldi, Mateusz Malinowski, Andrea Tacchetti, David Raposo, Adam Santoro, Ryan Faulkner, Çağlar Gülçehre, H. Francis Song, Andrew J. Ballard, Justin Gilmer, George E. Dahl, Ashish Vaswani, Kelsey R. Allen, Charles Nash, Victoria Langston, Chris Dyer, Nicolas Heess, Daan Wierstra, Pushmeet Kohli, Matthew M. Botvinick, Oriol Vinyals, Yujia Li, and Razvan Pascanu. Relational inductive biases, deep learning, and graph networks. *CoRR*, abs/1806.01261, 2018.
- [14] Jie Zhou, Ganqu Cui, Shengding Hu, Zhengyan Zhang, Cheng Yang, Zhiyuan Liu, Lifeng Wang, Changcheng Li, and Maosong Sun. Graph neural networks: A review of methods and applications. *AI Open*, 1:57–81, 2020.
- [15] Jacob Andreas. Measuring compositionality in representation learning. *CoRR*, abs/1902.07181, 2019.
- [16] Henry Brighton and Simon Kirby. Understanding linguistic evolution by visualizing the emergence of topographic mappings. *Artificial life*, 12(2):229–242, 2006.
- [17] Christian Bentz and Dimitrios Alikaniotis. The word entropy of natural languages. *arXiv preprint arXiv:1606.06996*, 2016.
- [18] Eugene Kharitonov, Rahma Chaabouni, Diane Bouchacourt, and Marco Baroni. Entropy minimization in emergent languages. In *Proceedings of the 37th International Conference on Machine Learning, ICML 2020, 13-18 July 2020, Virtual Event*, volume 119 of *Proceedings of Machine Learning Research*, pages 5220–5230. PMLR, 2020.

- [19] Noga Zaslavsky, Terry Regier, Naftali Tishby, and Charles Kemp. Semantic categories of artifacts and animals reflect efficient coding. *arXiv preprint arXiv:1905.04562*, 2019.
- [20] Eugene Kharitonov, Roberto Dessì, Rahma Chaabouni, Diane Bouchacourt, and Marco Baroni. EGG: a toolkit for research on Emergence of lanGuage in Games. <https://github.com/facebookresearch/EGG>, 2021.
- [21] Edward Gibson, Richard Futrell, Steven P Piantadosi, Isabelle Dautriche, Kyle Mahowald, Leon Bergen, and Roger Levy. How efficiency shapes human language. *Trends in cognitive sciences*, 23(5):389–407, 2019.
- [22] Matthias Fey and Jan Eric Lenssen. Fast graph representation learning with pytorch geometric. *CoRR*, abs/1903.02428, 2019.
- [23] Petar Veličković, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Liò, and Yoshua Bengio. Graph attention networks. *arXiv preprint arXiv:1710.10903*, 2018.
- [24] Shaked Brody, Uri Alon, and Eran Yahav. How attentive are graph attention networks? *CoRR*, abs/2105.14491, 2021.
- [25] Yunsheng Shi, Zhengjie Huang, Wenjin Wang, Hui Zhong, Shikun Feng, and Yu Sun. Masked label prediction: Unified message passing model for semi-supervised classification. *CoRR*, abs/2009.03509, 2020.